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(2) :- FIR Filter Design Using Frequency Sampling Method :-

- (i) Choose desired frequency response $H_d(e^{j\omega})$
- (ii) Take N samples of $H_d(e^{j\omega})$ to generate $H_d(k)$
- (iii) Take Inverse DFT to get $h(n)$

$$H_d(e^{j\omega}) \longrightarrow H_d(k) \longrightarrow h(n)$$

freq. Sampling

IDFT

$$\omega = \frac{2\pi k}{N}$$

$$\rightarrow h(n) = \frac{1}{N} \sum_{k=0}^{N-1} H(k) e^{jk n}$$

where $n = 0, 1, \dots, N-1$

$\rightarrow h(n)$ will have N samples. So it is FIR filter.

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Freq. Sampling Method

Type-I

$$\omega = 0$$

Type-II

$$\omega \neq 0$$

$N \rightarrow$ Order of filter.

Step I:-

$$\omega = \frac{2\pi k}{N} \quad k = 0, 1, \dots, N-1$$

Step II:-

$$H_d(k) = H_d(e^{j\omega}) \Big|_{\omega = \frac{2\pi k}{N}}$$

$$k = 0, 1, \dots, N-1$$

Step III:- Apply IDFT

$$h(n) = \frac{1}{N} \sum_{k=0}^{N-1} H_d(k) e^{jn\frac{2\pi k}{N}} \quad n = 0, 1, \dots, N-1$$

Now apply condition on ~~$h(n)$~~ $H_d(k)$ to be real valued, our formulas will be

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 $N = \text{odd}$:-

$$h(n) = \frac{1}{N} \left\{ H(0) + 2 \sum_{k=1}^{N-1/2} \operatorname{Re} \left\{ H(k) e^{j \frac{2\pi kn}{N}} \right\} \right\}$$

 $N = \text{even}$:-

$$h(n) = \frac{1}{N} \left[H(0) + 2 \sum_{k=1}^{\frac{N}{2}-1} \operatorname{Re} \left\{ H(k) e^{j \frac{2\pi kn}{N}} \right\} \right]$$

 $\therefore$ If $N = 4$

$$h(n) = \frac{1}{4} \left\{ H(0) + 2 \sum_{k=1}^1 \operatorname{Re} \left\{ H(k) e^{j \frac{2\pi kn}{4}} \right\} \right\}$$

(4)

Exp:-

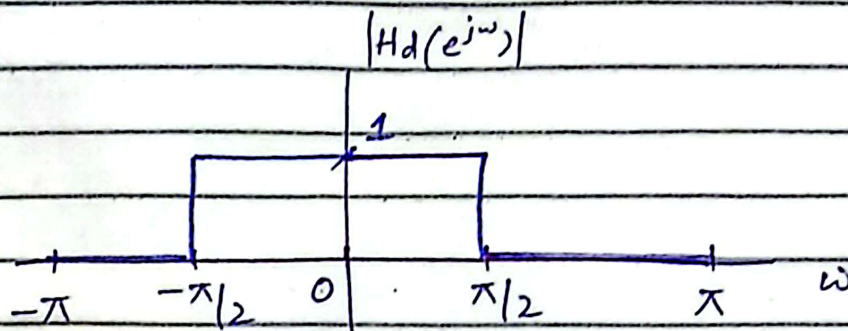
$$H_d(e^{j\omega}) = \begin{cases} e^{-j\frac{(M-1)\omega}{2}} & 0 \leq |\omega| \leq \pi/2 \\ 0 & \pi/2 \leq |\omega| \leq \pi \end{cases}$$

$M=7$ order of filter

Solution:-

Step-I:-

$$H_d(e^{j\omega}) = \begin{cases} e^{-j3\omega} & 0 \leq |\omega| \leq \pi/2 \\ 0 & \pi/2 \leq |\omega| \leq \pi \end{cases}$$



$$\omega = \frac{2\pi k}{N} = \frac{2\pi k}{7}$$

$$-\pi \leq \omega < \pi \Rightarrow k = 0$$

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Step II :- $H_d(e^{j\omega}) \rightarrow H(k)$

Sample $H_d(e^{j\omega})$.

put $\omega = \frac{2\pi k}{N}$; $k = 0, 1, \dots, N-1$

for $N=7 \Rightarrow \omega = \frac{2\pi k}{7}$; $k = 0, 1, \dots, 6$

$$H_d(e^{j\omega}) = \begin{cases} e^{-j\frac{6\pi k}{7}} & 0 \leq \frac{2\pi k}{7} \leq \pi/2 \\ 0 & \pi/2 < \frac{2\pi k}{7} \leq \pi \end{cases}$$

We have to find values of 'k'

$$\frac{2\pi k}{7} = \frac{\pi}{2} \Rightarrow k = 7/4$$

$$\frac{2\pi k}{7} = \pi \Rightarrow k = 7/2$$

$$\therefore H_d(k) = \begin{cases} e^{j\frac{6\pi k}{7}} & 0 \leq k \leq 7/4 \\ 0 & 7/4 \leq k \leq 7/2 \end{cases}$$

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Step III:- Obtain $h(n)$
 N is odd.

$$h(n) = \frac{1}{N} \left[H(0) + 2 \sum_{k=1}^{\frac{N-1}{2}} \operatorname{Re} \left\{ H(k) e^{j \frac{2\pi k n}{N}} \right\} \right]$$

$$\frac{N-1}{2} = \frac{7-1}{2} \Rightarrow (3) \quad H(0) = 1$$

$$h(n) = \frac{1}{7} \left[1 + 2 \sum_{k=1}^3 \operatorname{Re} \left\{ e^{-j \frac{6\pi k}{7}} e^{j \frac{2\pi k n}{7}} \right\} \right]$$

$$h(n) = \frac{1}{7} \left[1 + 2 \sum_{k=1}^3 \operatorname{Re} \left\{ e^{-j 2\pi k (3-n)/7} \right\} \right]$$

$$e^{-j\theta} = \underbrace{\cos \theta}_{\operatorname{Re}} - j \underbrace{\sin \theta}_{\operatorname{Im}}$$

$$\operatorname{Re} \left\{ e^{-j\theta} \right\} = \cos \theta$$

$$\therefore h(n) = \frac{1}{7} \left[1 + 2 \sum_{k=1}^3 \cos \left(\frac{2\pi k (3-n)}{7} \right) \right]$$

$$n = 0, 1, \dots, 6$$